

**FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION,
ISLAMABAD**

Annual Examination 2026

MATHEMATICS — Class IX

**Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation |
Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry**

Total Marks: 50

Time Allowed: 2 Hours 15 Minutes

Version: 1

General Instructions:

1. This paper has THREE sections: A, B and C. Attempt ALL questions from every section — there are no optional or "OR" questions anywhere on this paper.
2. Section A carries 10 MCQs (1 mark each). Circle the correct option letter in the Answer column.
3. Section B (6 short questions, 4 marks each) and Section C (2 long questions, 8 marks each) must be answered with complete working shown; answers without working may not receive full credit.
4. Use of calculators is not allowed. Write neatly and manage your time across all three sections.

SECTION A — Multiple Choice Questions (10 × 1 = 10 Marks)

#	Question	A	B	C	D	Answer
1	Simplify: $(3^4 \times 3^{-2}) \div 3^{-3} = ?$	3^3	3^5	3^9	3^{-1}	(A) (B) (C) (D)
2	$\sqrt{48} + \sqrt{27} = ?$	$5\sqrt{3}$	$7\sqrt{6}$	$7\sqrt{3}$	75	(A) (B) (C) (D)
3	$\log_5(1/25) = ?$	-2	2	1/2	-1/2	(A) (B) (C) (D)
4	If $\log 2 = 0.3010$, then $\log 128 = ?$	1.5050	1.8060	2.4080	2.1070	(A) (B) (C) (D)
5	The factorization of $x^2 - 14x + 49$ is:	$(x-7)(x+7)$	$(x-7)^2$	$(x+7)^2$	$(x-49)(x+1)$	(A) (B) (C) (D)
6	The HCF of $15a^3b^2$ and $20a^2b^3$ is:	$10a^2b^2$	$60a^3b^3$	$5a^2b^2$	$5a^3b^3$	(A) (B) (C) (D)
7	$ -8 + 3 = ?$	11	-11	5	-5	(A) (B) (C) (D)
8	If $-3 \leq 2x - 1 \leq 5$, then $x \in ?$	$[-2, 6]$	$[1, 3]$	$[-1, 6]$	$[-1, 3]$	(A) (B) (C) (D)
9	The distance between the points (0, 0) and (9, 12) is:	21	15	9	12	(A) (B) (C) (D)
10	The mid-point of the segment joining A(-5, 3) and B(1, -7) is:	$(-4, -4)$	$(2, 2)$	$(-2, -2)$	$(-2, 5)$	(A) (B) (C) (D)

SECTION B — Short Questions (6 × 4 = 24 Marks)

Attempt ALL 6 questions. No choice — every question is compulsory.

Q	Question	Marks
2(i)	Using the laws of exponents, simplify: (a) $16^{3/4} \times 16^{-1/4}$ (b) Simplify and write with positive exponents: $(x^{-2}y^3)^{-2} \div (x^3y^{-1})^2$	4
2(ii)	(a) Expand using the laws of logarithms: $\log[(p^3\sqrt{q})/r^4]$ (b) Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, evaluate $\log 0.9$.	4
2(iii)	The first polynomial is $P = x^2 + 2x - 24$. Its HCF with a second polynomial Q is $(x - 4)$, and their LCM is $x^3 + x^2 - 26x + 24$. Find the second polynomial Q.	4
2(iv)	Simplify to lowest form (all denominators non-zero): $1/(x^2-1) + 1/(x^2-3x+2) + 1/(x^2+3x+2)$	4
2(v)	(a) Solve for x and verify both solutions: $ 2x + 1 = 13$ (b) Solve the inequality $-7 < 3x + 2 \leq 14$, $x \in \mathbb{R}$, and write the solution set in set-builder notation.	4
2(vi)	Show that A(5, 7), B(2, 3) and C(10, -3) are the vertices of a right-angled triangle. State clearly at which vertex the right angle lies.	4

SECTION C — Long Questions (2 × 8 = 16 Marks)**Attempt BOTH questions. No choice — every question is compulsory.**

Q	Question	Marks
3(i)	(a) Reduce the following product of algebraic fractions to its lowest form (all denominators non-zero): $[(x^2-25)/(x^2+8x+15)] \times [(x^2+3x)/(x^2-3x-10)]$ [4] (b) Find the square root of $9x^4 - 12x^3 + 10x^2 - 4x + 1$ by the division method. [4]	8
3(ii)	Three survey pillars along a straight canal are located at P(1, 2), Q(4, 6) and R(7, 10), with coordinates measured in kilometres. (a) Find the distances PQ, QR and PR using the distance formula. [4] (b) Hence show that P, Q and R are collinear points. [2] (c) Find the mid-point of PR and show that it coincides with Q. [2]	8

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Unit 1 (Real Numbers) | Unit 2 (Logarithms) | Unit 4 (Factorization & Algebraic Manipulation) | Unit 5 (Linear Equations & Inequalities) | Unit 7 (Coordinate Geometry)**— End of Question Paper —**